TEST & LOAD REPORT

RoadWave 👋

End-to-end & load test report

Target: **https://www.getroadwave.com** · Generated 2026-04-29 · Repository commit **9aa59e6**

28 / 28	200 OK	Blocked
PLAYWRIGHT TESTS PASSING	PRODUCTION HEALTH (SINGLE REQUEST)	LOAD TEST VS. PROD (VERCEL WAF)

1. Scope

This report covers two distinct test campaigns run against the live RoadWave production site at `https://www.getroadwave.com` on 2026-04-29:

1. **Playwright end-to-end suite** — 28 functional tests covering every public page, the interactive demo (including manual chat input), per-campground demo routing, owner auth gates, public forms, and a concurrent-signup load check.
2. **k6 load test** — targeting 1000 concurrent VUs across three scenarios (scan QR, sign up, wave). Scaled down to 50 VUs for safety; observed Vercel edge-WAF behavior.

2. Playwright suite — 28/28 passing

All 28 tests passed in 23.3s, run from `tests/roadwave.test.js` against the live site. Tests deliberately avoid form submissions that would write real rows or trigger Resend emails; signup flows are exercised at the field-presence level only.

GROUP	TESTS	STATUS	NOTES
Public marketing pages	7	PASS	Homepage hero + CTAs; About; Contact (mailto resolves to `hello@getroadwave.com`); Privacy; Terms; Campgrounds page (lead form + nav); Tour splash.
Demo — interactive	8	PASS	Welcome heading on first paint; "See Nearby Campers" navigates correctly; all 7 nav tabs reachable; Wave button transitions card; Privacy mode switch reflected on Home; Meetups Hosted+Camper sections; Waves seed; **manual chat input** in a Crossed-paths conversation (typed message, asserted bubble appears).
Per-campground slug routing	4	PASS	`/demo/riverbend-rv-park`, `/oak-hollow-rv-resort`, `/pine-lake-campground`, `/coastal-pines-campground` — each renders its campground name correctly.
Owner auth gates	5	PASS	`/owner`, `/owner/dashboard`, `/owner/preview` all redirect anonymous traffic to `/owner/login`; login form has email + password; signup form has all four required fields.
Concurrent signups	1	PASS	5 parallel browser contexts hit `/owner/signup` simultaneously — the route handles parallel cold load without 5xx-ing.
Public forms (read-only)	2	PASS	Homepage "Request RoadWave at My Campground" form fields; campgrounds lead-form fields. Submission deliberately not exercised.
Total	**28**	PASS	23.3s wall time, 4 workers, retain-on-failure traces enabled.

3. Failures encountered & fixed during the suite

The first run produced 24/28; four failures were investigated and resolved:

TEST	ROOT CAUSE	FIX
Privacy tab can switch the mode badge	The mode buttons mix emoji + label + description; the accessible name starts with the emoji, so `name: /^Quiet/i` never matched.	Switched to `locator('button:has-text("Quiet")')` — reliably finds the Quiet button regardless of accessible-name prefix.
/owner/login renders email + password inputs	`<label>` elements in the form lack `htmlFor`, so Playwright's `getByLabel` can't bind them to inputs.	Switched the inputs to `locator('input[name="email"]')` / `name="password"` — matches the actual DOM attribute.
/owner/signup renders all four required fields	Same `htmlFor` issue as above.	Same fix: name-attribute selectors for `display_name`, `email`, `password`, `campground_name`.
5 concurrent visits to /owner/signup	Same `getByLabel` issue, applied across 5 parallel contexts.	Same fix: name-attribute selector.

4. k6 load test — 1000-VU plan vs. real-world result

A k6 script (`tests/load/roadwave-load.js`) was written to drive 1000 concurrent virtual users through three scenarios: *scan QR / load demo*, *sign up*, and *wave at another user*. The script is committed to the repo and parameterised so scenario / VU count / duration / target URL are env-flag driven.

> **Two production-side blockers prevented running 1000 VUs against the live site.** Both turn out to be correct production behavior — not bugs to fix.

4.1 Vercel edge-WAF blocks automated load traffic expected

All requests from k6 (with both default and browser user-agents) returned **HTTP 403** with a 31 KB attack-challenge response body. Vercel's edge filter detects automated load test traffic via TLS fingerprint, missing client hints, header pattern, and rate signals — not just the User-Agent string.

RUN	VUS	TARGET	RESULT
scan_only smoke	50	prod	403 100% of 3,077 requests blocked at edge
scan_only with browser UA	50	prod	403 Same result — Vercel uses other signals
Single curl-style request	1	prod	200 Origin healthy at normal volume

This is the *correct* behavior. Real load tests for Vercel-hosted apps run against staging environments with WAF rules relaxed, or from infrastructure Vercel allow-lists. Pointing 1000 destructive VUs at the live origin would have created 1000 real `auth.users` rows in Supabase and blown past the Resend email rate limit immediately — even if the WAF hadn't blocked it.

4.2 Local dev server isn't load-testable expected

Falling back to `npm run dev` (Next.js dev server) on localhost, the same scan-only scenario at 50 VUs collapsed: 70 requests over 70 seconds (~1 req/s actual throughput, vs the 50 req/s the script attempted), with 42% timeouts and successful requests averaging 2 seconds. This is expected: the dev server is single-threaded, runs HMR overhead per request, and uses unminified bundles — it isn't designed for concurrent load.

4.3 Latency (low-volume, prod, single requests)

Latency at normal call rates against production is excellent:

- Mean response time: **51 ms**
- p95 response time: **79 ms**
- p99 response time: **198 ms**
- TLS handshake & connect overhead: typically <2 ms (kept-alive in the test)

These numbers reflect what a real guest sees scanning a QR code — well under the Web Vitals "good" threshold for TTFB (200 ms).

5. Failures fixed in this push

Adjacent to the test run, these fixes were applied and shipped:

- `/owners` (plural) was a 404. Created a server-component redirect to `/campgrounds`.
- Campgrounds hero paragraph reworded twice for cleaner wording around interest matching.
- Personal Gmail removed from the codebase — replaced with `hello@getroadwave.com` across legal pages, owner panel, footer, lead-email NOTIFY_TO, owner QR fallback. Footer Contact link points at the new `/contact` page.
- New `/contact` page with mailto CTA — matches the rest of the marketing chrome.
- Demo Waves tab pre-seeded with mock data + a sample-conversation preview so the mechanic is visible on first load.
- Demo Home gained a prominent "See Nearby Campers" CTA below the action tile grid.
- Profile `years_rving` validator + DB constraint relaxed from 100 to 9999 (was surfacing "Too big: expected number to be <=100" to users).
- RLS migration `0015_profiles_self_insert_update.sql` re-asserts INSERT and UPDATE policies on `profiles`, fixing avatar save errors of the form "new row violates row-level security policy".
- Sign Out is now visible on every owner page — including the previously-uncovered `/owner/preview` — and the guest sign-out redirects to the marketing homepage.

6. Current site status

SURFACE	STATUS	NOTES
Marketing pages (`/`, `/about`, `/contact`, `/campgrounds`, `/privacy`, `/terms`, `/tour`)	Healthy	All pass Playwright smoke; 51ms avg latency.

SURFACE	STATUS	NOTES
Demo (`/demo` and `/demo/[slug]`)	Healthy	Interactive flows verified incl. chat input and tab nav.
Owner login / signup pages	Healthy	Both render, gate works correctly for anonymous traffic.
Owner dashboard (auth gates)	Healthy	Anonymous traffic to dashboard / preview redirects to login as expected.
Owner authenticated flows	Not tested	Would require real owner credentials; out of scope for this run.
Vercel edge protection	Active	Blocks automated load test traffic at the edge — confirmed working as intended.
Pending DB migrations	Apply when ready	0011 (enum), 0012 (RLS), 0013 (messages), 0014 (years_rving), 0015 (profile self-write). Application-level fallbacks are in place; applying the migrations stops the warning logs.

7. Recommendations

1. **Run pending migrations** 0011-0015 in Supabase Studio to clear application-level fallback paths and warning log noise.
2. **Add a staging Vercel project** with WAF rules relaxed (or env-tuned) for true load testing — production should never be load-tested directly.
3. **Wire dedicated `/api/load/signup` and `/api/load/wave` endpoints** for k6 to drive write paths in staging without the PKCE / server-action complexity.
4. **Add htmlFor to login / signup form labels** for screen-reader parity and to enable Playwright's `getByLabel` in future tests.
5. **Set up a scheduled Playwright run** (GitHub Actions, cron-style) to catch regressions on the marketing surface and demo flows before they ship.

Generated by `scripts/generate-test-report.mjs` via Playwright headless Chromium. Source: `tests/roadwave.test.js` (Playwright suite) and `tests/load/roadwave-load.js` (k6 script). Repository: github.com/goroadwave/RoadWave.